Information Security

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Motivation


إِنَّ اللَّهَ مَعَ الصَّابِرِينَ
"Indeed Allah is with those who patiently endure."
[Qur'an, 8:46]

Classical Encryption Techniques

1. Substitution Techniques

2. Transposition Techniques

1. Substitution Techniques

Letters are replaced by other letters or symbols or numbers

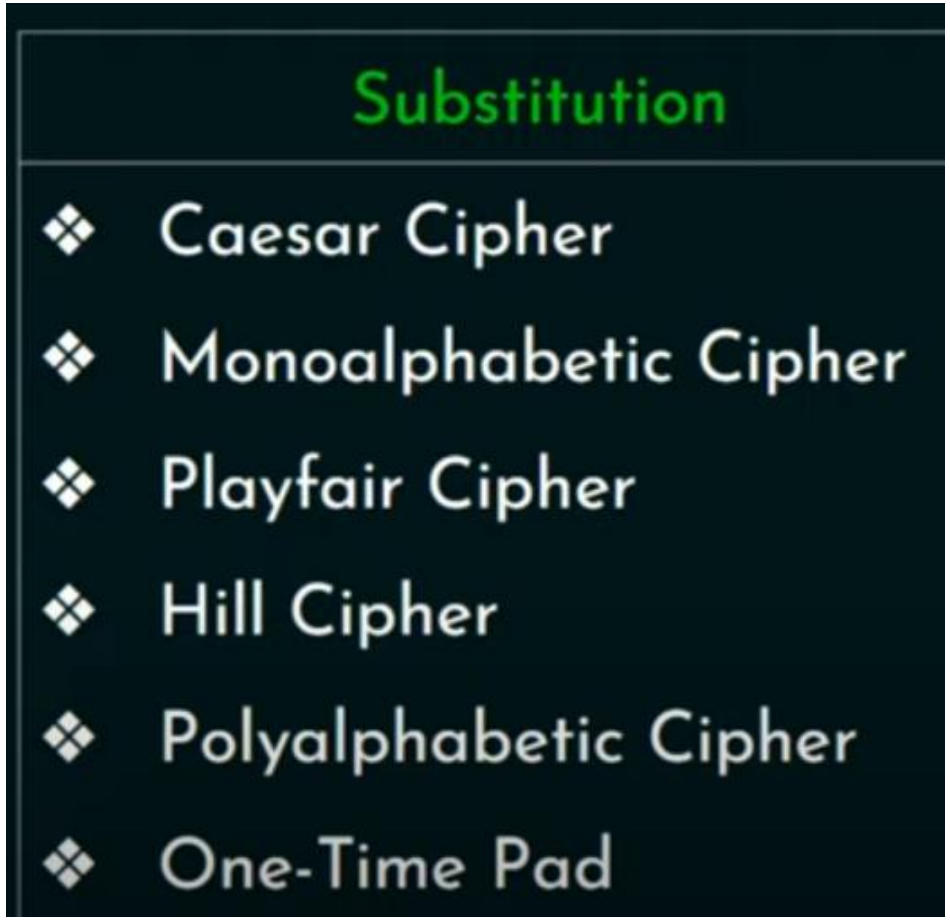
a	b	c	d	e	f	g	h	i	j	k	l	m
n	o	p	q	r	s	t	u	v	w	x	y	z

a → M
b → X
x → Z
g → A

Key information known to both sender and receiver

Plain Text: bag
Cipher Text: XMA

1. Substitution Techniques



1. Substitution Techniques

1.1. Poly Alphabetic Ciphers

1.2. One-Time-Pad

1. Substitution Techniques

1.1. Poly Alphabetic Ciphers

Polyalphabetic Cipher

★ To improve on the simple monoalphabetic technique.

★ General name: Polyalphabetic substitution cipher.

Common features

1. A set of related monoalphabetic substitution rules is used.
2. A key determines which particular rule is chosen for a given transformation.

1.1. Poly Alphabetic Ciphers (Vigenere Cipher)

Vigenere Cipher

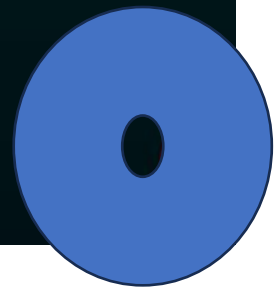
★ It consists of the 26 Caesar ciphers with shifts of 0 through 25.

Encryption process:

$$C_i = (P_i + K_{i \bmod m}) \bmod 26$$

Decryption process:

$$P_i = (C_i - K_{i \bmod m}) \bmod 26$$



Encryption of Vigenere Cipher

Cipher Text = Plain Text + Key

Cipher Text = w + d

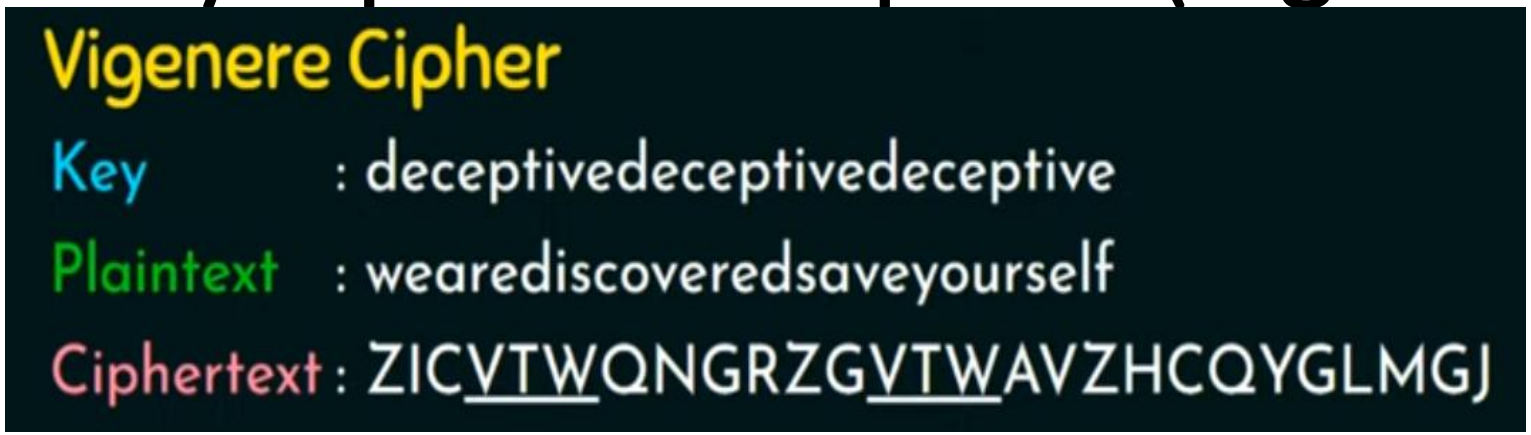
Cipher Text = 22 + 3

Cipher Text = 25

Cipher Text = Z

Now use this procedure for the whole encryption process.

1.1. Poly Alphabetic Ciphers (Vigenere Cipher)



A 0	B 1	C 2	D 3	E 4	F 5	G 6	H 7
I 8	J 9	K 10	L 11	M 12	N 13	O 14	P 15
Q 16	R 17	S 18	T 19	U 20	V 21	W 22	X 23
Y 24	Z 25						

Key	3 d	4 e	2	4	15	19	8	21	4	3	4	2	4
PT	22 w	4 e	0	17	4	3	8	18	2	14	21	4	17
CT	25	8	2	21	19	22	16	13	6	17	25	6	21

Decryption of Vigenere Ciphere

Plain Text= Cipher Text-Key

Plain Text= Z – d

Plain Text= 25 – 3

Plain Text= 22

Plain Text= w

Now use this whole procedure for Decryption

Note: If we get –ve value then add 26 in it.

Decryption of Vigenere Cipher

Vigenere Cipher

Key : deceptivedeceptivedeceptive

Plaintext : wearediscoveredsaveyourself

Ciphertext : ZICVTWQNGRZGVTWAVZHCQYGLMGJ

A 0	B 1	C 2	D 3	E 4	F 5	G 6	H 7
I 8	J 9	K 10	L 11	M 12	N 13	O 14	P 15
Q 16	R 17	S 18	T 19	U 20	V 21	W 22	X 23
Y 24	Z 25						

CT	25	8	2	21	19	22	16	13	6	17	25	6	21
Key	3	4	2	4	15	19	8	21	4	3	4	2	4
PT	22	4	0	17	4	3	8	18	2	14	21	4	17

Decryption of Vigenere Cipher

Plain text = We are discovered save yourself

1.1. Poly Alphabetic Ciphers (Autokey System)

Autokey system

★ The periodic nature of the keyword can be eliminated by using a non-repeating keyword that is as long as the message itself.

★ Vigenère proposed autokey system, in which a keyword is concatenated with the plaintext itself to provide a running key.

Example

Key : deceptivewearediscoveredsav

Plaintext : wearediscoveredsaveyourself

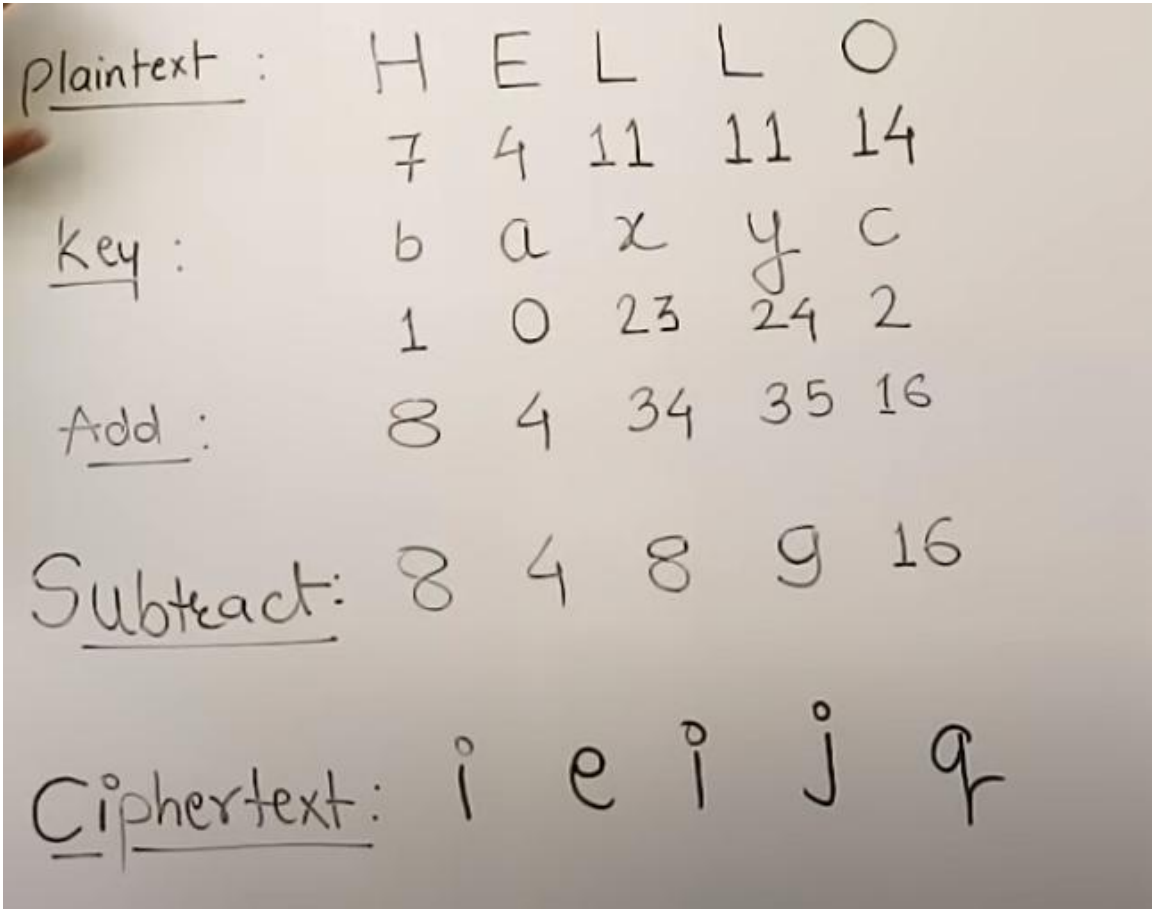
Ciphertext : ZICVTWQNGKZEIIGASXSTSLVVWLA

Q#1: Apply Vigenere Cipher for encryption and decryption on the plain text (attack at down) using Key (road).

Plaintext	a	t	t	a	c	k	a	t	d	a	w	n
Key	r	o	a	d	r	o	a	d	r	o	a	d
Ciphertext	r	h	t	d	t	y	a	w	u	o	w	q

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Example: Encryption



34-26= 8
35-26= 9

Decryption

Cipher Text: i e i j q

8 4 8 9 14

Key: b a x y c

1 0 23 24 2

---

7 4 -15 -15 12

+26 +26

---

7 4 11 11 12

Plain Text: H E L L O

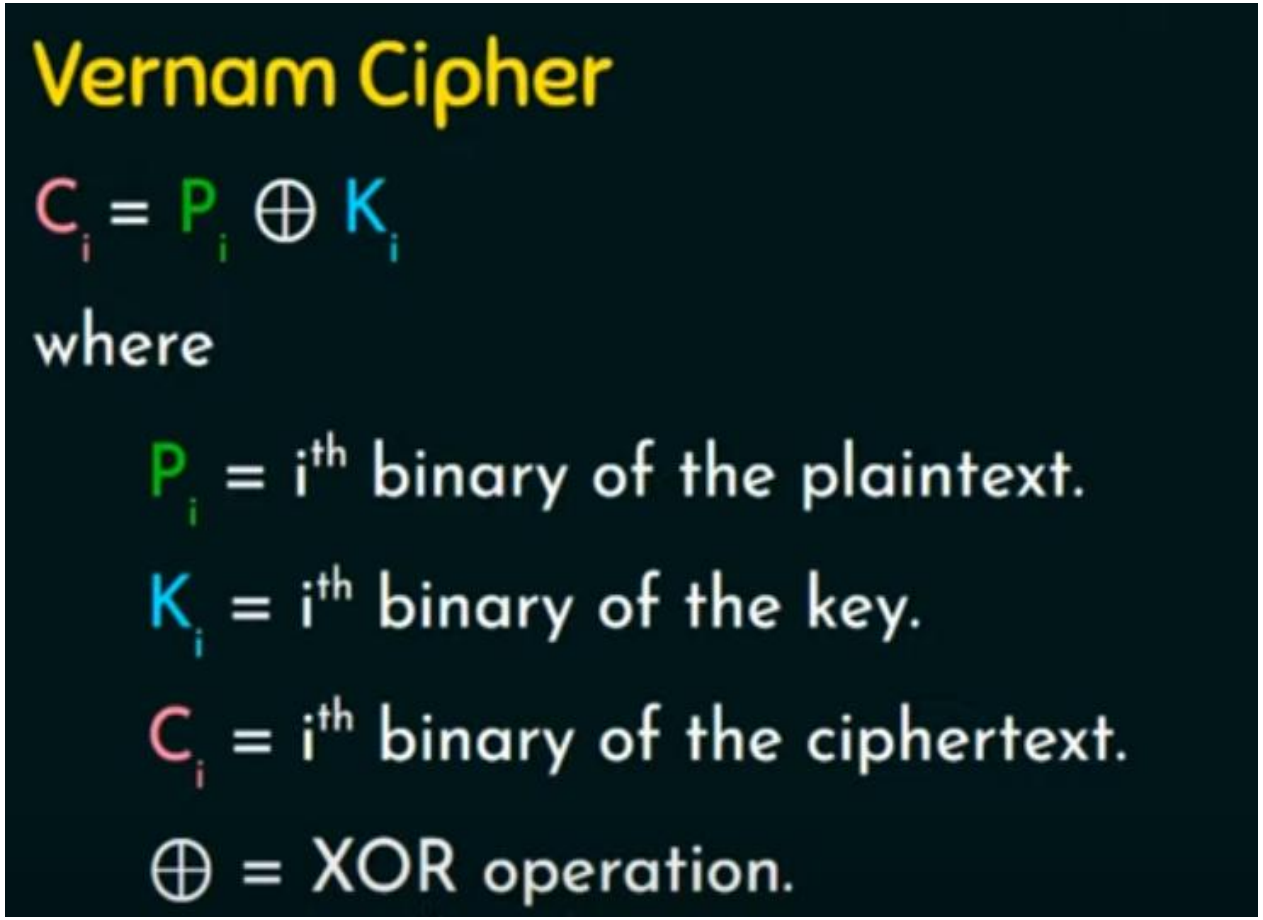
1.1. Poly Alphabetic Ciphers (Vernam Ciphers)

Vernam Cipher

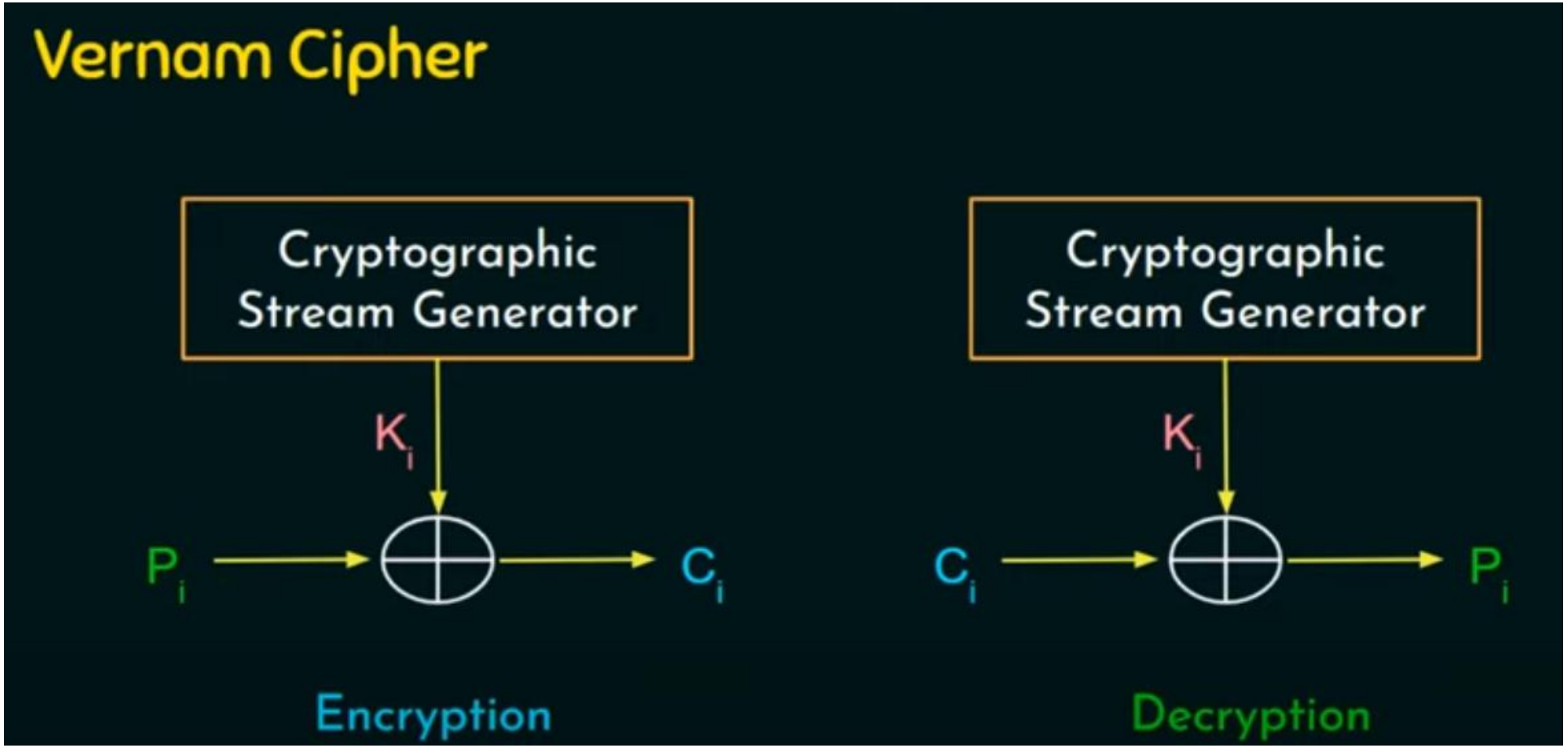
- ★ Need of ultimate defense against cryptanalytic attack.
- ★ Length of the keyword = Length of the plaintext.
- ★ No statistical relationship to it.
- ★ AT&T engineer named Gilbert Vernam in 1918.
- ★ His system works on binary bits rather than letter.
- ★ The system can be expressed as follows:

$$C_i = P_i \oplus K_i$$

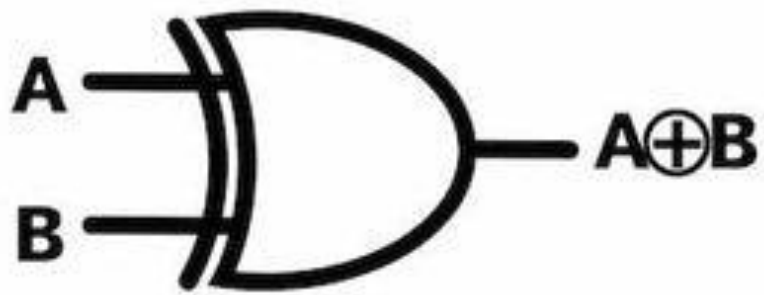
1.1. Poly Alphabetic Ciphers (Vernam Ciphers)



1.1. Poly Alphabetic Ciphers (Vernam Ciphers)



XOR (Exclusive OR)



*eXclusive OR

2 input XOR gate

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

1.1. Poly Alphabetic Ciphers (Vernam Ciphers)

Example-01: Let

Plain Text= A= 0001, Key= G=1111, F=1110

To Encrypt

PT	A	0	0	0	1
Key	G	1	1	1	1
CT XOR	F	1	1	1	0

To Decrypt

CT XOR	F	1	1	1	0
Key	G	1	1	1	1
PT	A	0	0	0	1

Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII
0	00000000	000	00	NUL	32	00100000	040	20	SP	64	01000000	100	40	@	96	01100000	140	60	`
1	00000001	001	01	SOH	33	00100001	041	21	!	65	01000001	101	41	A	97	01100001	141	61	a
2	00000010	002	02	STX	34	00100010	042	22	"	66	01000010	102	42	B	98	01100010	142	62	b
3	00000011	003	03	ETX	35	00100011	043	23	#	67	01000011	103	43	C	99	01100011	143	63	c
4	00000100	004	04	EOT	36	00100100	044	24	$	68	01000100	104	44	D	100	01100100	144	64	d
5	00000101	005	05	ENQ	37	00100101	045	25	%	69	01000101	105	45	E	101	01100101	145	65	e
6	00000110	006	06	ACK	38	00100110	046	26	&	70	01000110	106	46	F	102	01100110	146	66	f
7	00000111	007	07	BEL	39	00100111	047	27	'	71	01000111	107	47	G	103	01100111	147	67	g
8	00001000	010	08	BS	40	00101000	050	28	(	72	01001000	110	48	H	104	01101000	150	68	h
9	00001001	011	09	HT	41	00101001	051	29	)	73	01001001	111	49	I	105	01101001	151	69	i
10	00001010	012	0A	LF	42	00101010	052	2A	*	74	01001010	112	4A	J	106	01101010	152	6A	j
11	00001011	013	0B	VT	43	00101011	053	2B	+	75	01001011	113	4B	K	107	01101011	153	6B	k
12	00001100	014	0C	FF	44	00101100	054	2C	,	76	01001100	114	4C	L	108	01101100	154	6C	l
13	00001101	015	0D	CR	45	00101101	055	2D	-	77	01001101	115	4D	M	109	01101101	155	6D	m
14	00001110	016	0E	SO	46	00101110	056	2E	.	78	01001110	116	4E	N	110	01101110	156	6E	n
15	00001111	017	0F	SI	47	00101111	057	2F	/	79	01001111	117	4F	O	111	01101111	157	6F	o
16	00010000	020	10	DLE	48	00110000	060	30	0	80	01010000	120	50	P	112	01110000	160	70	p
17	00010001	021	11	DC1	49	00110001	061	31	1	81	01010001	121	51	Q	113	01110001	161	71	q
18	00010010	022	12	DC2	50	00110010	062	32	2	82	01010010	122	52	R	114	01110010	162	72	r
19	00010011	023	13	DC3	51	00110011	063	33	3	83	01010011	123	53	S	115	01110011	163	73	s
20	00010100	024	14	DC4	52	00110100	064	34	4	84	01010100	124	54	T	116	01110100	164	74	t
21	00010101	025	15	NAK	53	00110101	065	35	5	85	01010101	125	55	U	117	01110101	165	75	u
22	00010110	026	16	SYN	54	00110110	066	36	6	86	01010110	126	56	V	118	01110110	166	76	v
23	00010111	027	17	ETB	55	00110111	067	37	7	87	01010111	127	57	W	119	01110111	167	77	w
24	00011000	030	18	CAN	56	00111000	070	38	8	88	01011000	130	58	X	120	01111000	170	78	x
25	00011001	031	19	EM	57	00111001	071	39	9	89	01011001	131	59	Y	121	01111001	171	79	y
26	00011010	032	1A	SUB	58	00111010	072	3A	:	90	01011010	132	5A	Z	122	01111010	172	7A	z
27	00011011	033	1B	ESC	59	00111011	073	3B	;	91	01011011	133	5B	[	123	01111011	173	7B	{
28	00011100	034	1C	FS	60	00111100	074	3C	<	92	01011100	134	5C	\	124	01111100	174	7C	|
29	00011101	035	1D	GS	61	00111101	075	3D	=	93	01011101	135	5D	]	125	01111101	175	7D	}
30	00011110	036	1E	RS	62	00111110	076	3E	>	94	01011110	136	5E	^	126	01111110	176	7E	~
31	00011111	037	1F	US	63	00111111	077	3F	?	95	01011111	137	5F	_	127	01111111	177	7F	DEL

Encryption

1.1. Poly Alphabetic Ciphers (Vernam Ciphers)

Example-02: q2z #w>

PT	R	0	1	0	1	0	0	1	0
Key	#	0	0	1	0	0	0	1	1
CT	q	0	1	1	1	0	0	0	1

Plain Text: RED
Key: #w>
Cipher Text: q2z

PT	E	0	1	0	0	0	1	0	1
Key	w	0	1	1	1	0	1	1	1
CT	2	0	0	1	1	0	0	1	0

PT	D	0	1	0	0	0	1	0	0
Key	>	0	0	1	1	1	1	1	0
CT	z	0	1	1	1	1	0	1	0

Decryption

1.1. Poly Alphabetic Ciphers (Vernam Ciphers)

Example-02: q2z #w>

CT	q	0	1	1	1	0	0	0	1
Key	#	0	0	1	0	0	0	1	1
PT	R	0	1	0	1	0	0	1	0

CT	2	0	0	1	1	0	0	1	0
Key	w	0	1	1	1	0	1	1	1
PT	E	0	1	0	0	0	1	0	1

CT	z	0	1	1	1	1	0	1	0
Key	>	0	0	1	1	1	1	1	0
PT	D	0	1	0	0	0	1	0	0

Cipher Text: q2z

Key: #w>

Plain Text: RED

Example-03

A **0**	B **1**	C **2**	D **3**	E **4**	F **5**	G **6**	H **7**
I **8**	J **9**	K **10**	L **11**	M **12**	N **13**	O **14**	P **15**
Q **16**	R **17**	S **18**	T **19**	U **20**	V **21**	W **22**	X **23**
Y **24**	Z **25**						

Encryption:
Plain Text: H E L L O
Key: N C B T A
Cipher Text: ??????

H E L L O
7 4 11 11 14

N C B T A
13 2 1 19 0 +

---

20 6 12 30 14
-26

---

20 6 12 04 14
U G M E O

Cipher Text: U G M E O

Example-03

A **0**	B **1**	C **2**	D **3**	E **4**	F **5**	G **6**	H **7**
I **8**	J **9**	K **10**	L **11**	M **12**	N **13**	O **14**	P **15**
Q **16**	R **17**	S **18**	T **19**	U **20**	V **21**	W **22**	X **23**
Y **24**	Z **25**						

Decryption:

Cipher Text: U G M E O

Key: N C B T A

Plain Text: **?????????**

U G M E O

20 6 12 4 14

N C B T A

13 2 1 19 0 -

---

7 4 11 -15 14

+26

---

7 4 11 11 14

H E L L O

Plain Text: **H E L L O**

Rules

- >25 then -26
- Negative value then + 26

1.1. Poly Alphabetic Ciphers (Vernam Ciphers)

Vernam Cipher – Cryptanalysis

- ★ Construction of the key.
- ★ The use of a running loop of tape that eventually repeated the key.
- ★ The system worked with a very long but repeating keyword.
- ★ This technique can be broken with sufficient ciphertext, the use of known or probable plaintext sequences, or both.

Jazak Allah Khair